Jyotishka Ray Choudhury

Curriculum Vitae

ISyE Main 439, Georgia Tech
755 Ferst Dr NW, Atlanta
GA 30332, United States
jchoudhury3@gatech.edu
jyotishkarc.github.io

Education

2024 – Present **Doctor of Philosophy [Ph.D.] in Machine Learning**, ***School of Industrial & Systems Engineering (ISyE), Georgia Institute of Technology***, Atlanta, GA, United States
- **Advisors: Prof. Ashwin Pananjady** and **Prof. Yao Xie**
- Awarded the *ISyE Beerman President's Fellowship*
- Awarded *Ronald J. and Carol T. Beerman Fellowship* for the first year in the program

2022 – 2024 **Master of Statistics [M.Stat]**, ***Indian Statistical Institute***, Kolkata, WB, India
- First Division (*Theoretical Statistics* Specialization)

2019 – 2022 **Bachelor of Statistics [B.Stat] (Hons.)**, ***Indian Statistical Institute***, Kolkata, WB, India
- First Division with Distinction (Highest honors)
- Awarded the *INSPIRE Scholarship*

2007 – 2019 **Secondary & Higher Secondary Education**, ***Ramakrishna Mission Boys' Home High School***, Rahara, Khardaha, WB, India

Broad Areas of Interest

- **Statistics:** High-dimensional statistics, Nonparametric statistics, Change-point detection, Conditional independence testing, Inference under missing data, Analysis of dependent data.
- **Machine Learning:** Statistical learning theory, Classification, Clustering, Bandits, Reinforcement learning, Generative models.
- **Optimization:** Stochastic optimization, Distributionally robust optimization.
- **Computing:** High-performance distributed computing on CPU, GPU, and clusters.

Publications & Preprints

(∗ denotes equal contribution)

❖ Published Papers:

[3] Basu, S.*, **Ray Choudhury, J.*** and Paul, D. and Das, S. (2025). *Dirichlet Process-Based Robust Clustering Using the Median-of-Means Estimator.* In: International Joint Conference on Artificial Intelligence (IJCAI), Main Track (pp. 4770–4778). URL: https://doi.org/10.24963/ijcai.2025/531.

[2] **Ray Choudhury, J.***, Saha, A.*, Roy, S., Dutta, S. (2023). *Robust Classification of High-Dimensional Data Using Data-Adaptive Energy Distance.* In: European Conference on Machine Learning and Knowledge Discovery in Databases: Research Track (ECML-PKDD) (pp. 86–101). Lecture Notes in Computer Science, vol 14173. Springer, Cham. URL: https://doi.org/10.1007/978-3-031-43424-2_6.

[1] Roy, S., **Ray Choudhury, J.**, Dutta, S. (2022). *On Some Fast And Robust Classifiers For High Dimension, Low Sample Size Data.* In: International Conference on Artificial Intelligence and Statistics (AISTATS) (pp. 9943–9968). PMLR, Volume 151. URL: https://proceedings.mlr.press/v151/roy22a.html.

❖ Preprints (Unpublished/Submitted):

[2] Niu, Z., **Ray Choudhury, J.** and Katsevich, E. (2024). *The conditional saddlepoint approximation for fast and accurate large-scale hypothesis testing.* arXiv preprint. URL: https://arxiv.org/abs/2407.08911. Submitted to Biometrika.

[1] Niu, Z., **Ray Choudhury, J.** and Katsevich, E. (2024). *The saddlepoint approximation for averages of conditionally independent random variables.* arXiv preprint. URL: https://arxiv.org/abs/2407.08915.

Academic Accolades

August, 2024 Awarded the **ISyE Beerman President's Fellowship** and **Ronald J. and Carol T. Beerman Fellowship** by the **School of Industrial & Systems Engineering, Georgia Tech**, on admission to the Ph.D. program in Machine Learning.

January, 2022 Selected for the **Summer Research Fellowship 2022** by **Indian National Science Academy (INSA)**, at the Indian Institute of Science (IISc), Bangalore, India.

August, 2019 Awarded the **INSPIRE Scholarship** by the **Department of Science and Technology, Government of India**, for securing a rank within top 1%-ile in the West Bengal Higher Secondary Examination, 2019.

June, 2019 Qualified the ***Bachelor of Statistics (B.Stat) Entrance Examination*** of **Indian Statistical Institute (ISI)** with an **All India Rank 17**.

June, 2019 Qualified the **Bachelor of Science (B.Sc)** in Math+CS Entrance Examination of the **Chennai Mathematical Institute (CMI)**, and awarded the *Sriram scholarship*.

Pre-doctoral Research Projects & Internships

Summer 2023 **Saddlepoint Approximation to the distilled Conditional Randomization Test**

- **Supervisor:** *Prof. Eugene Katsevich*, University of Pennsylvania, PA, USA

Summer 2022 **Effect of Bitcoin Energy Consumption on Electricity Spot Prices using Nonparametric Multivariate Time Series Regression**

- **Supervisors:** *Prof. Soudeep Deb*, IIM Bangalore, India; *Prof. Rishideep Roy*, University of Essex, England; *Prof. Sayar Karmakar*, University of Florida, USA

Summer 2022 Summer 2021 **Robust Classification of High-Dimensional, Low Sample Size Data**

- **Supervisor:** *Prof. Subhajit Dutta*, Indian Institute of Technology (IIT) Kanpur, India

Winter 2020 **Dirichlet Process-based Automatic Clustering using Median-of-Means**

- **Supervisor:** *Prof. Swagatam Das*, *ECSU*, Indian Statistical Institute, Kolkata, India

Reading Projects

Spring 2023 **Introduction to Multi-Armed Bandits: Algorithms & Statistical Theory**

- **Supervisor:** *Prof. Pan Xu*, Duke University, NC, USA

Technical / Programming Skills

Computing **R** (Advanced), **Python** (Intermediate), **C++** (Intermediate)

Markup LaTeX, **Markdown**, **Rmarkdown**

Web App **Shiny** in R, **HTML**, **CSS**

Shell Script **Bash**

Teaching Experiences & Other Activites

Teaching Graduate Teaching Assistant (GTA) for ISYE-6416 and ISYE-7416 in Fall 2025.

Mentoring Trained and mentored a number of students in mathematics, for the undergraduate entrance examinations of the *Indian Statistical Institute* and other colleges from 2020 to 2023, both in online and offline modes.

Leadership Localization Head of the **Mathematical Talent Reward Programme** (MTRP), 2023, an examination on advanced high school mathematics, organized by the students of ISI Kolkata.

Co-curricular Reading, Table Tennis, Football (Soccer), Cooking.

Additional Information

Languages	**Bengali** (Mother tongue), **English** (Professional), **Hindi** (Fluent), **Spanish** (Introductory)
Interests	Learning about science, technology, history, wildlife, languages, etc.
Homepage	**https://jyotishkarc.github.io**
Email	jyotishka.stat@gmail.com
ORCID	**0000-0002-7070-683X**
Scholar	https://scholar.google.com/citations?user=NhTxu3MAAAAJ&hl=en
GitHub	https://github.com/jyotishkarc